

Plaza1 / Plaza2 Case-Study Showcase Manual

AI-ready implementation guide for comparative SLAM posterior recovery using NF-iSAM, Slices, and Smoothed Slices

Purpose of this manual

This file tells an AI/developer exactly how to build the Plaza1 and Plaza2 showcase inside the MATLAB App. The objective is not only to run algorithms; it is to demonstrate posterior recovery: samples, marginals, modes, uncertainty shape, runtime, and incremental behavior. The comparative methods are: (1) NF-iSAM, (2) Slices Perspective, and (3) Smoothed Slices / Recursive Conditional-Smoothing Slices (our proposed extension).

Critical distinction

The Slices paper reports Plaza2 results. The NF-iSAM paper reports Plaza1 and Plaza2 results. Therefore, for Plaza2 the showcase should aim to reproduce paper-style Slices and NF-iSAM comparisons. For Plaza1, NF-iSAM is the direct paper reference, while Slices and Smoothed Slices are comparative extensions implemented in the same experimental framework.

1. Source basis and what must be reproduced

Source	Relevant implementation facts to preserve
Slices paper	Uses factor graph variable elimination. At each step it defines $P_{\text{joint}}(\theta_{\text{j}}, S_{\text{j}} D_{\text{j}})$, obtains $f_{\text{new}}(S_{\text{j}} D_{\text{j}})$ by marginalizing θ_{j} , approximates it with Monte Carlo slices, estimates conditionals, then performs backward-pass marginal recovery. It introduces Lemma 1 for sample generation without unary factors and an MMD early-stopping heuristic for backward pass.
NF-iSAM paper	Uses Bayes-tree clique conditionals. Learns normalizing flows for clique densities $p(F_{\text{C}}, S_{\text{C}} z_{\text{C}})$, extracts separator density $p(S_{\text{C}} z_{\text{C}})$ and conditional $p(F_{\text{C}} S_{\text{C}}, z_{\text{C}})$, updates only changed subtrees, and samples top-down through learned conditionals.
Smoothed Slices / RCS formulation note	Our extension: reinterpret nested Slices sampling as a tower of conditional expectations. Replace repeated deep sample trees, where useful, by cached conditional smoothing surfaces $R_{\text{r}}(x_{\text{i}}, s)$. It does not learn a full posterior density; it learns expected future factor contributions.

Non-negotiable implementation principle

The app must show all three methods side-by-side on the same dataset, same measurement stream, same random seed policy, same plotting conventions, same metric computation, and same incremental time indices. Differences in performance must come from the inference methods, not from different preprocessing, plotting, or evaluation code.

2. Required working-directory hierarchy

The AI/developer must assume this project structure

The screenshots show a root Project folder with Instructions, Literature, presentation, and the current Smoothed Slices formulation PDF. The implementation should preserve that organization and add Data, src, results, and figures folders exactly as specified below.

Project/

├─ Instructions/

| └─ nf_isam_exact_implementation.pdf

| └─ slices_perspective_exact_implementation.pdf

- | |─ smoothed_slices_exact_implementation.pdf
- | |─ smoothed_slices_matlab_app_spec.pdf
- | |─ plaza_case_study_showcase_manual.docx <-- this file
- |
- |─ Literature/
 - | |─ A Slices Perspective for Incremental Nonparametric Inference...
 - | |─ Incremental_Non-Gaussian_Inference_for_SLAM_Using_Normalizing_Flows...
 - | |─ Probabilistic_data_association_for_semantic_SLAM...
 - | |─ Batch Continuous-Time Trajectory Estimation...
 - | |─ other supporting papers
- |
- |─ Data/
 - | |─ Plaza1/
 - | | |─ raw/
 - | | |─ processed/
 - | | |─ metadata/
 - | |─ Plaza2/
 - | | |─ raw/
 - | | |─ processed/
 - | | |─ metadata/
- |
- |─ src/
 - | |─ app/ # MATLAB App Designer files
 - | |─ common/ # factor graph, parser, metrics, plotting
 - | |─ methods/
 - | | |─ nf_isam/
 - | | |─ slices/
 - | | |─ smoothed_slices/
 - | |─ experiments/
 - | | |─ run_plaza1_showcase.m
 - | | |─ run_plaza2_showcase.m
- |
- |─ cache/
- | |─ nf_isam/

```

|   └─ slices/
|   └─ smoothed_slices/
|
└─ results/
    └─ <timestamp>_<dataset>_<scenario>/
        └─ config.json
        └─ metrics.csv
        └─ method_outputs.mat
        └─ figures/
        └─ diagnostics/
        └─ exported_app_snapshot/
|
└─ figures/
    └─ paper_style/
    └─ diagnostics/
    └─ animations/

```

Folder	Responsibility
Instructions	Implementation specifications only. The AI should read these before coding. Do not write algorithm outputs here.
Literature	Reference papers and supporting articles. Use for understanding algorithms and plots; do not modify.
Data	Raw and processed Plaza1 / Plaza2 data. Processed files must be deterministic and reproducible from raw files.
src	All source code. Each method must expose the same backend API so the app can compare them.
cache	Reusable method-specific artifacts: Bayes tree flows, Slices factors, Smoothed Slices surfaces.
results	Every run creates a timestamped folder with config, metrics, .mat outputs, figures, diagnostics, and app snapshots.
figures	Global saved figures for presentation use. Do not depend on MATLAB figure windows only.

3. Plaza1 and Plaza2 datasets: showcase meaning

Dataset	What the papers say / what to demonstrate	Showcase role
Plaza1	NF-iSAM reports that Plaza1 and Plaza2 are real range-only datasets available in the GTSAM software distribution; each sequence has odometry and range measurements for a planar vehicle with four unknown landmarks. The paper uses calibrated range distances by fitting and subtracting an affine distance-bias model.	Use as the primary NF-iSAM-style real-data case. Apply all three methods to show whether Slices and Smoothed Slices can recover comparable trajectory, landmark, and posterior-shape behavior.
Plaza2	The Slices paper uses Plaza2: odometry and range measurements to four unknown landmarks, nonlinear measurements and non-Gaussian likelihoods. It reports incremental runtime/RMSE and marginal distributions for landmark 2 over time.	Use as the main paper-aligned comparative case for all three methods. It is the key showcase because both NF-iSAM and Slices papers provide paper-style expectations.

Dataset loader requirement

The AI must implement one common parser API for both datasets:

```
dataset = loadPlazaDataset(datasetName, rootDir, options).
```

It must output poses, odometry increments, range measurements, landmark IDs when available, ground truth when available, timestamps / step indices, and calibration metadata.

All three methods must consume the same processed dataset structure.

Required processed dataset structure:

```
dataset.name           % 'Plaza1' or 'Plaza2'
dataset.steps          % vector of incremental step indices
dataset.odom           % table/struct: fromPose, toPose, delta_x, delta_y(optional), delta_theta, covariance
dataset.ranges         % table/struct: poseId, landmarkId or candidates, measuredRange, sigma, timestamp
dataset.landmarkIds    % list of landmark IDs, expected length 4 for the Plaza sequences used in the papers
dataset.gt.poses       % ground-truth pose trajectory if available
dataset.gt.landmarks   % ground-truth landmarks if available
dataset.calibration    % affine range bias correction model and residual noise model
dataset.scenarios      % known association / ambiguous association / ADA fraction settings
```

4. Preprocessing and measurement models

Preprocessing rule

All preprocessing must be method-independent. The app must not allow NF-iSAM, Slices, or Smoothed Slices to use different calibrations, noise models, data-association candidates, or ground-truth alignment unless explicitly marked as an ablation.

Known range factor: $\text{ilde } r_i^w = \|t_i^w - l_j^w\|_2 + \xi_i, \text{quad } \xi_i \sim \text{mathcal N}(0, \sigma^2).$

Unknown association factor: $p(\text{ilde } r_i \mid t_i^w, \text{mathcal L}_i) = \frac{1}{|\text{mathcal L}_i|} \sum_{l_j^w \in \text{mathcal L}_i} p(\text{ilde } r_i \mid t_i^w, l_j^w).$

Range calibration: $r_{\text{cal}} = r_{\text{raw}} - (a, r_{\text{raw}} + b) \text{quad } \text{text{or the equivalent affine model fitted from calibration data}}.$

Preprocessing step	Implementation instruction
Range calibration	Implement a calibration module that can fit or load the affine range-bias model used for Plaza datasets. Store fitted coefficients and residual statistics. If calibration data are unavailable, allow a config switch to load supplied coefficients or run uncalibrated with a warning.
Pose alignment	For trajectory error plots, implement optional Kabsch-Umeyama alignment before comparing estimated and ground-truth trajectories, matching the NF-iSAM paper style.
Data association modes	Support known association and configurable ambiguous data association (ADA). For ADA, wipe landmark IDs for selected range factors and assign candidate set L_i to all landmarks or to a configured subset.
Noise model	Use common odometry and range noise parameters across all methods. Store covariance/sigma in config.json and display them in the app.
Incremental stream	Convert data to an ordered list of factors added at each time step. Each method receives the same factor-addition events.

5. Shared backend API for the three methods

Why this matters

The showcase is comparative. The app must call each method through the same interface. This makes the plots, metrics, cache snapshots, and step sliders comparable.

Required MATLAB method interface:

```

method = createMethod(methodName, config)

method.initialize(dataset)

for k = 1:numSteps

    resultStep = method.update(dataset.events{k})

    methodState{k} = method.getStateSnapshot()

    posterior{k} = method.getPosteriorSamples(config.numPosteriorSamples)

    marginals{k} = method.getMarginals(config.queryVariables)

    diagnostics{k} = method.getDiagnostics()

end

methodResults = method.finalize()

```

Every method must return:

- posterior samples for poses and landmarks
- marginal samples / marginal density estimate for selected variables
- estimated trajectory mean and landmark mean
- runtime split: preprocessing, update, inference, sampling, plotting
- method-specific diagnostics
- cache state needed for slider playback

Method	Required internal diagnostics
NF-iSAM	Bayes tree, changed subtree, clique list, frontal/separators variables, training samples per clique, flow loss history, flow training time, posterior sampling time, number of trainable parameters if available.
Slices	Elimination order, removed factors $F_{\{j-1\}}(\theta_{\{j\}})$, separator S_j , selected sampling factor f_{prime} , number of samples N_j , f_{new} summaries, conditionals, backward-pass MMD values, early-stop variable/time.
Smoothed Slices	Everything from Slices plus smoothing path $xi_0 \rightarrow \dots \rightarrow xi_H$, surface R_r cardinalities, active successor sizes $ N_r $, surface approximation error proxy, surface-cache hits, proposal-guided sampling diagnostics.

6. Method-specific case-study implementation

6.1 NF-iSAM showcase implementation

$$p(\Theta | z) = \prod_{C \in \mathcal{C}} p(F_C | z_C, S_C, z_C).$$

$$p(F_C, S_C | z_C) \propto \prod_{C \in \mathcal{C}_C} p(S_q | z_q) \prod f_i(\Theta_{\{f_i\}}).$$

- Build or update the Bayes tree using the selected elimination ordering.
- For each changed clique C , generate training samples from an intermediate density using efficient ancestral sampling. Treat selected loop-closing / measurement variables as unobserved during sample simulation when following the NF-iSAM strategy.
- Train a triangular normalizing flow for the clique density $p(S_C, F_C | z_C)$.
- Fix observations, partition the learned map into separator-density and clique-conditional components, cache samplers $p(S_C | z_C)$ and $p(F_C | S_C, z_C)$.
- During incremental updates, retrain flows only in the changed subtree. Reuse flows outside the subtree. Posterior sampling still traverses the full Bayes tree top-down.

NF-iSAM paper-style defaults

Use rational-quadratic spline autoregressive flows when available. Recommended paper-style defaults: two-layer fully connected conditioner networks, hidden width 8, number of RQ spline pieces $K=9$, default training samples 2000, and stop training when the relative change between the average loss over the latest 50 iterations and the previous 50 iterations drops below 1%.

6.2 Slices showcase implementation

$$P_{\text{joint}}(\theta_j, S_j \mid D_j) = \eta_j^{-1} \prod_{f_i \in F_{j-1}} f_i(\theta_j) f_i(\Theta_i).$$

$$P_{\text{joint}}(\theta_j, S_j \mid D_j) = P(\theta_j \mid S_j, D_j) f_{\text{new}}(S_j \mid D_j).$$

$$f_{\text{new}}(S_j \mid D_j) = \eta_j^{-1} \int \prod_{f_i \in F_{j-1}} f_i(\theta_j) f_i(\Theta_i) d\theta_j.$$

$$\hat{f}_{\text{new}}(S_j \mid D_j) = \frac{\eta_j^{-1}}{N_j} \sum_{n=1}^{N_j} \prod_{f_i \in F_{j-1}} f_i(\theta_j^{(n)}) f_i(\Theta_i \setminus \theta_j^{(n)}).$$

- Use one consistent elimination order. For Plaza2, reproduce the Slices paper style by eliminating landmark variables last during affected re-elimination.
- At each forward-elimination step, record the reduced graph G_{j-1} , eliminated variable θ_j , removed factor set $F_{j-1}(\theta_j)$, separator S_j , selected sampling factor f_{prime} , generated samples $\theta_j^{(n)}$, approximate f_{new} , and conditional estimator.
- If θ_j has no unary factor, apply Lemma-1 sampling logic: sample from the factor created by a previous elimination that still contains θ_j .
- During backward pass, compute marginals as mixtures of conditional slices using samples from $P_{\text{hat}}(S_j \mid D)$.
- Implement MMD early stopping in the backward pass: compare new marginal samples to cached previous marginal samples; stop when MMD is below threshold.

Slices Plaza2 settings to expose in UI

Use $N=150$ samples per step as the paper-style Slices setting. For early stopping, expose $N_M=100$ MMD samples and threshold $\text{vartheta}=1e-4$ as defaults. These must be editable in the app.

6.3 Smoothed Slices showcase implementation

Status of this method

Smoothed Slices is our proposed extension. The app must label it clearly as Proposed / Research Branch, not as an author-specified algorithm from the Slices paper.

$$R_r(\mathbf{x}_{r,s}) = Z_r(\mathbf{x}_r) \mathbb{E}_{\mathbf{x}_{r+1} \sim q_r(\cdot \mid \mathbf{x}_r)} [a_r(\mathbf{x}_{r+1}, s) R_{r+1}(\mathbf{x}_{r+1}, s)].$$

$$\hat{f}_{\text{new}, \text{RCS}}(s) = \frac{c_0}{N_0} \sum_{n=1}^{N_0} b_0(\mathbf{x}_0^{(n)}, s) \hat{R}_0(\mathbf{x}_0^{(n)}, s).$$

$$\text{nested cost} \sim N^H | \mathcal{S}|, \quad \text{RCS sparse cost} \sim \sum_{r=0}^{H-1} |X_r| | \mathcal{S}|.$$

- Start from the same Slices elimination structure, same data stream, same separators, and same local factor products.
- Whenever the Slices implementation would generate nested samples along a Lemma-1 sampling path, identify the path $\mathbf{x}_0 \rightarrow \mathbf{x}_1 \rightarrow \dots \rightarrow \mathbf{x}_H$ and the final separator s .
- Construct conditional smoothing surfaces $R_r(\mathbf{x}_r, s)$ over finite support spaces X_r and separator support S .
- Use sparse active successor sets $N_r(a)$ instead of dense all-to-all propagation when possible.
- Cache R_r surfaces across increments. On a new measurement, update only affected surfaces and reuse all others.
- If additional samples are needed, use the future surface R_{r+1} to guide proposals, but preserve the Slices structural proposal through a defensive mixture.

7. Plaza1 / Plaza2 experiment protocols

Experiment	Dataset	Scenario	Methods	Primary outputs
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P1-A	Plaza1	Known association or native association as loaded from data.	NF-iSAM, Slices, Smoothed Slices	Trajectory, landmarks, posterior samples, marginal KDEs, runtime, RMSE.
P1-B	Plaza1	Configurable ADA fractions. Recommended initial sweep: 0%, 20%, 40%, 60% if parser/generator supports it; mark exact fractions as configurable.	NF-iSAM, Slices, Smoothed Slices	Multimodal posterior recovery, data-association ambiguity robustness, MMD/RMSE/runtime.
P2-A	Plaza2	Paper-style incremental range-only SLAM with four landmarks.	NF-iSAM, Slices, Smoothed Slices	Paper-style Plaza2 plots: marginal of landmark 2 across time, RMSE/time over steps.
P2-B	Plaza2	Stress-test sample counts and early-stop thresholds.	Slices, Smoothed Slices; optional NF-iSAM	Sensitivity of posterior recovery to N, N_M, vartheta, surface cardinalities $ X_r $, $ S $.

Plaza2 landmark-2 plot requirement

For the Slices-style Plaza2 showcase, create marginal posterior plots for landmark L2 over separate x and y axes at the time steps where it receives important range measurements in the paper-style narrative: time steps 0, 8, and 22. These plots must compare NF-iSAM, Slices, and Smoothed Slices on the same axes.

8. Figure suite to generate

Figure family	Required content	Why it matters
Map / trajectory	2-D map with ground truth trajectory, odometry trajectory, estimated mean trajectory, landmark means, uncertainty ellipses or sample clouds.	Shows classical SLAM recovery and qualitative correctness.
Posterior samples	Scatter plot of posterior samples at selected time steps. Use consistent colors per method. Show ambiguous modes when present.	Demonstrates full posterior recovery, not only point estimates.
Marginal KDEs	For selected poses/landmarks, plot x and y marginal densities. For Plaza2, include L2 at time steps 0, 8, 22.	Matches paper-style posterior-shape visualization.
RMSE over incremental step	RMSE of trajectory / landmarks versus time step for all methods.	Performance comparison with classical paper metrics.
Runtime over incremental step	Total update runtime and stacked runtime components: preprocessing, forward/upward, training/smoothing, backward/downward, sampling, plotting.	Distinguishes online feasibility.
MMD / distributional distance	Joint MMD and selected marginal MMD versus time if reference samples are available. If no reference, use pairwise method-to-method MMD and mark it as relative.	Evaluates distribution shape, not just mean.
Slices process panel	Variable elimination slider: $G_{[j-1]}$, $\theta_{[j]}$, removed factors, separator S_j , selected sampling factor, f_{new} , conditional, generated samples.	Documents the Slices mechanism visually.
NF-iSAM process panel	Bayes-tree slider: current clique, frontal/separator split, changed subtree, flow training loss, learned separator density, conditional sampler.	Documents the NF-iSAM workflow.
Smoothed Slices process panel	Surface-recursion slider: ξ path, R_r surface heatmap/table, $ X_r $, $ N_r $, $ S $, cached vs updated surfaces, nested vs sparse cost.	Shows the novel method and its online cardinality advantage.
Diagnostics dashboard	Effective sample size, MMD early-stop values, flow losses, cache hits, surface approximation residuals, sample count per method.	Explains why each method succeeds or fails.

Figure export requirement

Every figure shown in the app must be exportable by a single Generate Figures / Save Diagnostics button. The button must save PNG, PDF/SVG where supported, MATLAB .fig, and a .mat snapshot of the data used to draw the figure. All axes labels, legends, titles, textboxes, and annotations must use MATLAB LaTeX interpreter when supported.

Required figure export callback:

```
function onGenerateFiguresButtonPushed(app,event)
```

```

runId = app.CurrentRunId;

outDir = fullfile(app.ProjectRoot,'results',runId,'figures');

diagDir = fullfile(app.ProjectRoot,'results',runId,'diagnostics');

ensureDir(outDir); ensureDir(diagDir);

figHandles = app.collectFigureHandles();      % UIAxes and detached figures

for i = 1:numel(figHandles)

    exportgraphics(figHandles{i}, fullfile(outDir, sprintf('fig_%02d.png',i)), 'Resolution', 300);

    exportgraphics(figHandles{i}, fullfile(outDir, sprintf('fig_%02d.pdf',i)), 'ContentType', 'vector');

end

save(fullfile(diagDir,'diagnostics_snapshot.mat'), 'appState','methodResults','metrics','config');

writetable(metrics, fullfile(diagDir,'metrics.csv'));

writeConfigJson(config, fullfile(diagDir,'config.json'));

end

```

9. MATLAB App panels and user interaction

Panel / Tab	Required controls	Required visual output
Dataset Setup	Project root picker; dataset selector Plaza1/Plaza2; scenario selector known/ambiguous; sample seed; noise settings; calibration toggle.	Loaded-data summary, number of poses, ranges, landmarks, candidate associations, ground-truth availability.
Run Control	Run All Methods; Run Selected Method; Step Forward; Reset; Save Figures/Diagnostics; Export Run Package.	Progress bar, current step, current method, runtime status, warnings.
Map & Posterior	Time-step slider; method checkboxes; variable selector.	Map, trajectory, samples, landmarks, range arcs/circles, uncertainty ellipses.
Marginals	Variable selector; coordinate selector x/y/theta; time-step selector.	KDE/marginal curves for all methods; optional analytic/reference curve if available.
Metrics	Metric selector RMSE/MMD/runtime/ESS/cache hits.	Time-series plots and final summary table.
Slices Process	Elimination-step slider j; toggle factor graph/Bayes net; show equations.	$G_{j-1} \rightarrow G_j$ visualization, θ_{j-1} , S_j , $F_{j-1}(\theta_{j-1})$, f_{prime} , f_{new} , conditional.
NF-iSAM Process	Clique slider C; subtree toggle; training-iteration slider.	Bayes tree, changed subtree, clique factor graph, flow loss, separator/frontal split, conditional sampler.
Smoothed Slices Process	Path-level slider r; surface selector R_r ; cardinality toggle.	R_r surface heatmaps, active successor sets, cost comparison, cache status.
Diagnostics Export	Generate all figures; save diagnostics; create report package.	Run folder tree preview and export log.

LaTeX interpreter rule

All app text that MATLAB supports must use Interpreter="latex": titles, axis labels, legends, annotations, UI labels where possible, and equation panels. If a MATLAB UI component does not support LaTeX, render the equation into a UIAxes or use formatted HTML only as a fallback, and document the fallback.

10. Metrics and diagnostics definitions

$$\mathrm{RMSE}_{\text{pose}}(k) = \sqrt{\frac{1}{N_x(k)} \sum_i \| \hat{x}_i(k) - x_i^{\text{gt}} \|^2}.$$

$$\mathrm{MMD}^2(X,Y) = \frac{1}{n^2} \sum_{i,j} k(x_i, x_j) + \frac{1}{m^2} \sum_{i,j} k(y_i, y_j) - \frac{2}{nm} \sum_{i,j} k(x_i, y_j).$$

$$ESS = \frac{(\sum_n w_n)^2}{\sum_n w_n^2}.$$

Metric	How to compute	Use in showcase
RMSE	Compare posterior-sample empirical mean or method mean to aligned ground truth.	Classical accuracy trend; easy to interpret.
MMD	Use samples from method and reference distribution; if no true reference, compare methods pairwise and label as relative MMD.	Posterior-shape accuracy; important for non-Gaussian/multimodal cases.
Runtime	Measure wall-clock with tic/toc around preprocessing, update, training/smoothing, backward pass, posterior sampling, plotting.	Online feasibility.
Flow loss	NF-iSAM only: training loss per clique and relative loss-change stopping condition.	Shows training stability and cost.
MMD early-stop value	Slices/Smoothed Slices: value between new and cached marginal in backward pass.	Explains early stopping and missed/accepted updates.
Surface cardinality	Smoothed Slices: $ X_r $, $ N_r $, $ S $, storage $ X_r S $, sparse cost $ X_r N_r S $.	Shows why the method may reduce nested computation.
ESS / weight degeneracy	For sampling-based estimators, compute ESS at relevant factors/variables.	Shows proposal quality and when sampling fails.

11. Step-by-step showcase execution protocol

1. Open the MATLAB App from Project/src/app/SmoothedSlicesDemoApp.mlapp.
2. Select Project root. The app must automatically verify Instructions, Literature, Data, src, cache, results, and figures folders.
3. Select Plaza1 or Plaza2. The app loads or builds the processed dataset and displays pose/range/landmark counts.
4. Select scenario: native associations, known associations, or ambiguous-data-association stress test.
5. Select methods: NF-iSAM, Slices, Smoothed Slices. Default: all three.
6. Set common sample seed and common posterior sample count for visualization/evaluation.
7. Click Run All Methods. The app runs methods sequentially or in parallel, but metric timing must be method-isolated and fair.
8. During the run, update the process panels so the user can pause and inspect variable elimination, Bayes-tree clique processing, and smoothing-surface recursion.
9. After the run, move the time-step slider to reproduce paper-style snapshots: Plaza2 L2 marginals at time steps 0, 8, and 22; Plaza1/Plaza2 map snapshots at selected ambiguity-resolution steps.
10. Click Generate Figures / Save Diagnostics. Verify that figures, metrics.csv, config.json, and diagnostics_snapshot.mat are written to the run folder.
11. Open the generated figure folder and ensure all paper-style and diagnostic figures exist.

12. Acceptance checklist for the AI/developer

Category	Acceptance condition
Data	Plaza1 and Plaza2 load into one shared dataset struct. Raw and processed files are separated. Calibration metadata is saved.
Methods	All three methods run through the same method.update / getPosteriorSamples / getMarginals / getDiagnostics interface.
Fairness	All methods receive the same event stream, same ground truth, same noise config, and same query variables.
App UI	Tabs exist for dataset setup, run control, map/posterior, marginals, metrics, Slices process, NF-iSAM process, Smoothed Slices process, diagnostics export.
LaTeX	Every supported MATLAB title/label/legend/equation panel uses LaTeX interpreter.
Figures	Generate Figures button saves PNG and vector exports, .fig where possible, metrics.csv, config.json, and .mat diagnostic snapshot.
Paper-style plots	Plaza2 L2 marginal x/y plots at selected steps; RMSE and runtime over incremental steps; posterior sample maps.
Process sliders	User can drag through Slices elimination steps, NF-iSAM clique/subtree steps, and Smoothed Slices surface-recursion steps.
Early stopping	Slices and Smoothed Slices show MMD values, threshold, and stop decision per backward pass.
Smoothed Slices cardinality	The UI displays $ X_r $, $ N_r $, $ S $ and cost comparison for every surface update.

13. Concrete code modules to write

Minimum module list:

```

src/common/loadPlazaDataset.m

src/common/preprocessPlazaData.m

src/common/buildFactorEvents.m

src/common/computeMetrics.m

src/common/computeMMD.m

src/common/alignTrajectoryKabschUmeyama.m

src/common/plotPosteriorMap.m

src/common/plotMarginalKDEs.m

src/common/exportAllFiguresAndDiagnostics.m


src/methods/nf_isam/createNFISAMMethod.m

src/methods/nf_isam/updateBayesTree.m

src/methods/nf_isam/trainCliqueFlow.m

src/methods/nf_isam/extractCliqueConditionalSampler.m

src/methods/nf_isam/samplePosteriorTopDown.m


src/methods/slices/createSlicesMethod.m

src/methods/slices/eliminateVariableSlices.m

src/methods/slices/estimateNewFactorSlices.m

src/methods/slices/estimateConditionalSlices.m

src/methods/slices/backwardMarginalSlices.m

src/methods/slices/mmdEarlyStop.m


src/methods/smoothed_slices/createSmoothedSlicesMethod.m

src/methods/smoothed_slices/buildSmoothingPath.m

src/methods/smoothed_slices/updateConditionalSurface.m

src/methods/smoothed_slices/updateSparseSurfaceRecursion.m

src/methods/smoothed_slices/guideProposalWithSurface.m

src/methods/smoothed_slices/cacheSmoothingSurfaces.m


src/app/SmoothedSlicesDemoApp.mlapp

```

Implementation priority

First implement robust common plotting, data loading, metrics, and export. Then implement baseline method wrappers. The app

can initially use mock outputs for one method, but final acceptance requires real outputs for all three methods and identical comparison figures.

14. AI implementation prompt to include at the top of the coding task

You are implementing a MATLAB App Designer demonstration for comparative full-posterior SLAM inference on Plaza1 and Plaza2. Read all PDFs in Project/Instructions first, then use Project/Literature as algorithmic source material. Implement a single fair data pipeline and compare NF-iSAM, Slices Perspective, and Smoothed Slices. The purpose is to demonstrate posterior recovery, not only point estimates. All figures must use LaTeX interpreter where MATLAB supports it. The app must include process sliders for Slices variable elimination, NF-iSAM Bayes-tree clique processing, and Smoothed Slices recursive conditional-smoothing surfaces. The Generate Figures button must export every visible figure and all diagnostics. Do not silently change noise models, calibration, data association, or ground truth per method. Save every run into a timestamped results folder with config.json, metrics.csv, method_outputs.mat, figures, diagnostics, and app snapshots.

15. References encoded in this manual

- M. Shienman, O. Levy-Or, M. Kaess, and V. Indelman, "A Slices Perspective for Incremental Nonparametric Inference in High Dimensional State Spaces," IROS, 2024.
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